

Trace-Driven Evaluation of Post-Quantum Authentication for ADS-B Using OpenSky Network Data

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1. Introduction

2. Background & Related Work

2.1 ADS-B 1090ES and Authentication Constraints

Automatic Dependent Surveillance-Broadcast (ADS-B) is a cooperative surveillance system in which aircraft periodically broadcast their identity, position, velocity, and other state information without requiring interrogation from a ground station. For ADS-B operating on the 1090 MHz Extended Squitter (1090ES) data link, this information is transmitted using 112-bit Mode S Extended Squitter messages. Of these 112 bits, 56 constitute the ME field carrying the ADS-B application data, while the remaining fields are required for message identification, aircraft addressing, and parity/interrogator identification. The ME field is itself allocated according to the ADS-B message type and therefore does not constitute unused capacity available for arbitrary security material. Moreover, 1090ES provides no generic fragmentation and reassembly mechanism through which larger objects can be transported across multiple messages [1, 2, 3].

ADS-B transmissions are periodic and asynchronous, and share the 1090 MHz channel with other surveillance traffic. Depending on the information being transmitted and the operational state of the aircraft, multiple Extended Squitters may be generated per second. A representative modeling value of up to 6.2 ADS-B transmissions per second per aircraft corresponds to approximately 0.0744% of raw channel airtime per aircraft when an Extended Squitter duration of approximately 120 μ s is considered [4]. Although small for an individual aircraft, aggregate channel occupancy increases with aircraft density and with other Mode S and surveillance transmissions operating on the same frequency. Consequently, an authentication mechanism that introduces additional Extended Squitters also introduces additional channel load into an already shared communication resource.

The existing 1090ES protocol does not provide cryptographic source authentication or integrity protection. Its parity mechanisms provide error detection and support Mode S addressing functions, but do not establish the authenticity of the transmitter or protect message contents against deliberate modification or fabrication. The protocol furthermore provides no native mechanisms for transporting cryptographic credentials, establishing sessions, exchanging keys, or associating large authentication objects with sequences of ADS-B messages. Introducing digital signatures into 1090ES must therefore be treated as a protocol-extension problem rather than as the addition of a security field to an existing message.

For a signature or other authentication object of size S_{auth} bits, even the optimistic assumption that the complete 56-bit ME field of an additional Extended Squitter could be dedicated to authentication

yields a theoretical minimum fragment count of

$$n_{\text{frag,min}} = \left\lceil \frac{S_{\text{auth}}}{56} \right\rceil. \quad (1)$$

An implementation would require part of this capacity for functions such as identification, sequencing, message association, and replay protection, such that the actual number of transmissions would be greater. Fragmentation affects not only communication overhead but also the probability that a receiver obtains the complete authentication object. Under the simplifying assumption of independent fragment reception with probability p_{rx} , the probability of receiving all n required fragments is

$$P_{\text{complete}} = p_{\text{rx}}^n. \quad (2)$$

This relationship is not intended as a complete model of 1090 MHz reception, but illustrates why authentication schemes requiring many fragments become increasingly sensitive to message loss. Fragmentation additionally delays authentication until sufficient security material has been received and reconstructed.

These constraints are relevant because authentication must coexist with the surveillance function and a heterogeneous installed base of ADS-B equipment. Successive 1090ES standards have historically remained operational for extended periods, making prolonged coexistence between equipment generations a practical consideration [5, 6, 7, 1]. The feasibility of ADS-B authentication is therefore determined not only by cryptographic security or computational performance, but also by the amount of authentication material that must be conveyed over the channel and its interaction with observed traffic. The subsequent evaluation consequently considers authentication-object size, fragmentation and transmission overhead, additional channel occupancy, authentication latency, successful reconstruction of authentication material, and receiver verification workload.

2.2 Existing ADS-B Authentication Research

The absence of native authentication in ADS-B has motivated a body of research into mechanisms for establishing message integrity and source authenticity. Existing approaches include asymmetric digital signatures, identity-based and certificate-less cryptography, delayed-disclosure symmetric authentication, and physical- or infrastructure-based verification [8]. Despite their different security architectures, cryptographic proposals face a common challenge: authentication must be introduced within the restricted communication capacity of ADS-B while preserving its broadcast operation and compatibility with existing infrastructure.

Several works have investigated asymmetric authentication directly. Baek et al. proposed an on-line/offline identity-based signature scheme in which computationally expensive operations are shifted offline, reducing the online signing cost and avoiding dependence on conventional public-key certificates [9]. Other approaches have considered hierarchical identity-based signatures with batch verification and certificate-less short signatures, addressing receiver verification workload, key management, and communication overhead [10, 11]. An alternative line of research employs delayed key disclosure to avoid public-key operations on individual ADS-B messages. Prakash et al. proposed authentication through retroactive key publication [12], while other approaches have similarly investigated time-dependent symmetric authentication for the ADS-B broadcast environment.

More recently, the Compatible Authenticated Bandwidth-efficient Broadcast protocol for ADS-B (CABBA) combines the Timed Efficient Stream Loss-Tolerant Authentication (TESLA) protocol with

public-key infrastructure and the 8PSK phase-overlay capability introduced in DO-260C [13]. This allows authentication information to be conveyed without relying exclusively on additional conventional Extended Squitter transmissions. CABBA has been experimentally evaluated using software-defined radios, including its communication overhead and backward compatibility with legacy receivers [13].

Other research has considered physical- and infrastructure-based verification, including multilateration, time-difference-of-arrival, ranging, and direction-of-arrival techniques [14, 15]. These approaches can provide independent evidence concerning the physical origin or plausibility of a received transmission, but do not themselves provide cryptographic message-origin authentication and are therefore complementary to cryptographic approaches.

2.3 Post-Quantum Authentication and Research Gap

While ADS-B authentication has therefore received considerable attention, its treatment from a post-quantum perspective remains limited. Post-quantum cryptography has been investigated more broadly for aeronautical communications, particularly for security-aware future systems such as LDACS [16, 17, 18]. In contrast, comparatively little work has examined the integration of post-quantum authentication into the constrained 1090ES environment. Recent ADS-B research continues to investigate new signature constructions, including hierarchical identity-based schemes [19], but has focused primarily on cryptographic security and computational performance rather than the system-level interaction of authentication mechanisms with 1090ES traffic. This leaves a gap between the development and standardization of post-quantum digital signatures and their operational feasibility within ADS-B 1090ES. Rather than proposing a new authentication protocol, this work addresses this gap by evaluating how the size and computational characteristics of NIST-standardized and selected post-quantum digital signatures interact with observed ADS-B traffic. The analysis specifically considers fragmentation, communication overhead, channel occupancy, authentication latency, reconstruction success, and receiver verification workload under trace-derived operating conditions.

3. Evaluation Framework

3.1 Security Objective and Scope

The security objective considered in this work is the post-quantum cryptographic authentication of ADS-B surveillance messages, providing message-origin authentication and integrity against deliberate modification or fabrication. The evaluation therefore considers post-quantum digital signatures as a means of allowing a receiver to verify that surveillance information originates from the claimed aircraft and has not been modified after signing. Confidentiality is not considered, since ADS-B is inherently a broadcast surveillance service whose information is intended to be received by multiple surrounding users. The evaluation also does not address physical-layer threats such as jamming or establish whether the physical origin of a received signal corresponds to the position reported within the message. Such properties require complementary signal- or infrastructure-based verification mechanisms and are distinct from cryptographic message authentication.

The scope is limited to the system-level feasibility of integrating digital signatures into ADS-B rather than the design of a complete authentication protocol. The analysis considers the generation, transmission, reconstruction, and verification of authentication material and the resulting communication and computational requirements. Key provisioning, certificate distribution, trust-anchor management, revocation, and the detailed encoding of an eventual protocol extension are outside the scope. A valid association between an aircraft identity and the corresponding verification key is therefore assumed.

The evaluation is further restricted to ADS-B over 1090ES. Although Universal Access Transceiver (UAT) provides an alternative ADS-B data link, its operational deployment and communication architecture differ substantially from those of 1090ES. UAT operates on a dedicated 978 MHz channel and is deployed predominantly within the general-aviation environment and inside United States, whereas 1090ES provides the globally applicable ADS-B implementation and is required for higher-altitude operations in the United States [20, 21]. UAT also employs a different framing and transmission structure, with one aircraft transmission per one-second frame and substantially different message-capacity characteristics. Thus, extending the evaluation to UAT would require a separate link-specific authentication model rather than a direct application of the 1090ES model developed here. Restricting the analysis to 1090ES therefore provides a well-defined assessment of post-quantum authentication under the constraints of the globally deployed Mode S Extended Squitter environment, while the applicability of the methodology to other ADS-B data links remains a subject for future work.

3.2 Candidate Signature Selection

The evaluation considers post-quantum digital signature schemes that have reached an advanced stage of standardization. In response to the threat posed by quantum computers to widely deployed public-key cryptography, the U.S. National Institute of Standards and Technology (NIST) initiated an international standardization process to evaluate and standardize quantum-resistant public-key algorithms [22]. Candidate schemes underwent several rounds of public evaluation and cryptanalysis before a small number were selected for standardization. Restricting the present evaluation to schemes that have completed, or have been selected through, this process provides a well-defined set of comparatively mature post-quantum signatures whose specifications and security properties have received extensive external scrutiny. On this basis, ML-DSA and SLH-DSA, standardized as FIPS 204 and FIPS 205, respectively, and FN-DSA, selected by NIST for subsequent standardization as FIPS 206, are considered [23, 24, 25].

These three schemes also expose different characteristics that are directly relevant to ADS-B. ML-DSA is a lattice-based signature scheme providing comparatively balanced signature size and computational performance and represents NIST's principal standardized general-purpose post-quantum signature. FN-DSA, derived from FALCON, is also lattice-based but produces considerably smaller signatures, making it particularly relevant where communication capacity is restrictive. SLH-DSA is instead based on hash functions and therefore provides a structurally different post-quantum alternative, but produces substantially larger signatures. Evaluating these schemes consequently captures markedly different trade-offs between authentication-object size, computational performance, and cryptographic construction rather than comparing algorithms with broadly equivalent characteristics.

ECDSA is additionally included as a classical reference baseline for digital-signature-based authentication, rather than as a candidate for post-quantum authentication. ECDSA is a mature and standardized digital signature scheme and has also been specified for authentication within aeronautical communication architectures, including ECDSA with SHA-256 over the NIST P-256 curve for the Aeronautical Telecommunication Network using IPS [26, 27]. It is not, however, a native component of ADS-B 1090ES nor present in the OpenSky messages used in this study. P-256 is selected as the classical baseline because it provides approximately 128 bits of classical security while retaining a compact 64-byte signature when its two signature components are represented as fixed-width 256-bit integers. This provides a deliberately communication-efficient conventional reference against which the substantially larger authentication objects introduced by post-quantum signatures can be assessed. Since even such a compact conventional signature may exceed the capacity available within an individual 1090ES message, the baseline allows limitations inherent to introducing

Table 1. Digital signature schemes and parameter sets considered in the evaluation.

Role	Scheme	Parameter set	PK [B]	Signature [B]
Classical baseline	ECDSA	P-256	65	64
PQC candidate	ML-DSA	ML-DSA-44	1312	2420
PQC candidate	FN-DSA	FN-DSA-512	897	666
PQC candidate	SLH-DSA	SLH-DSA-SHA2-128s	32	7856

digital-signature authentication into ADS-B to be distinguished from those specifically aggravated by the transition to post-quantum cryptography. ECDSA is consequently subjected to the same fragmentation, transmission, and computational evaluation as the post-quantum candidates, but is not considered a viable long-term alternative to them due to its vulnerability to quantum attacks. The parameter sets and principal quantitative characteristics used in the evaluation are summarized in Table 1.

For each post-quantum scheme, the parameter set corresponding to NIST security category 1 is selected. This provides a consistent security level across the post-quantum candidates while representing the most communication-favorable parameterization of each scheme. Since the objective is to assess whether post-quantum signatures can be accommodated under the restrictive communication characteristics of 1090ES, selecting larger parameter sets would increase authentication overhead without providing a more favorable feasibility case. For SLH-DSA, the “s” parameterization is selected because it prioritizes smaller signatures over signing speed, consistent with the communication-constrained nature of the evaluated system.

3.3 Authentication Architectures

The integration of digital signatures into ADS-B requires distinguishing between the *authentication architecture* and the *transport of authentication material*. The authentication architecture determines which surveillance messages are covered by each signature, whereas the transport mechanism determines how the resulting signature is conveyed to the receiver. These are independent considerations: a signature may require fragmentation regardless of whether it authenticates a single ADS-B message or a group of messages. Fragmentation therefore does not constitute detached authentication in itself.

Two authentication architectures are considered: immediate authentication and detached authentication. Both assume that the receiver possesses an authenticated public key corresponding to the transmitting aircraft. They differ exclusively in the number of surveillance messages associated with each authentication event, and when those messages can be authenticated.

In the immediate architecture, every ADS-B message is authenticated independently. For each surveillance message m_i , the aircraft generates a dedicated signature

$$\sigma_i = \text{Sign}(sk, m_i), \quad (3)$$

where sk denotes the aircraft’s private signing key. Each message therefore has a one-to-one association with its corresponding signature. The surveillance message is transmitted according to the normal ADS-B transmission process, while the associated signature is conveyed as authentication material. If the signature cannot be accommodated within a single authentication transmission, it is fragmented across multiple transmissions and reconstructed by the receiver. The message becomes cryptographically authenticated only after its complete signature has been received, reconstructed where necessary, and successfully verified. The term *immediate* therefore refers to the generation

of an independent authentication event for every surveillance message, rather than implying that verification necessarily occurs immediately upon reception of that message.

In the detached architecture, a single signature authenticates a predetermined group of $k > 1$ ADS-B messages. The surveillance messages are transmitted normally and accumulated until the authentication interval k is reached. For a sequence $\{m_i, \dots, m_{i+k-1}\}$, a single signature is generated over the complete message sequence according to

$$\sigma_{i,k} = \text{Sign}(sk, m_i || m_{i+1} || \dots || m_{i+k-1}), \quad (4)$$

where $||$ denotes concatenation. The resulting signature is subsequently conveyed as authentication material and, as in the immediate architecture, may be transmitted as a single object or fragmented across multiple transmissions depending on its size. Once the receiver has obtained the required surveillance messages and the complete signature, the group can be verified and the constituent messages authenticated retrospectively. Detached authentication therefore reduces the frequency with which signatures must be generated and transmitted by amortizing one signature across multiple surveillance messages.

The authentication interval k provides a common representation of both architectures. Immediate authentication corresponds to $k = 1$, for which every surveillance message has its own signature, while detached authentication corresponds to $k > 1$, for which one signature covers multiple messages. The evaluation considers $k \in \{1, 5, 10, 20\}$, with $k = 1$ representing immediate authentication and $k \in \{5, 10, 20\}$ representing increasingly detached authentication intervals. Increasing k reduces the number of signatures generated and transmitted per surveillance message, but also increases the number of messages awaiting authentication and the time between reception of the first message in a group and completion of its authentication. The selected intervals are intended to expose this trade-off rather than prescribe operational authentication periods.

Authentication transport is modeled separately from this architectural distinction. For any value of k , the generated signature is treated as an authentication object of size S_{sig} . Where this object exceeds the capacity allocated to authentication material within an individual transmission, it is divided into multiple fragments that must be received and reconstructed before signature verification can occur. Consequently, signature size and fragmentation determine the communication cost of an individual authentication event, while the authentication interval k determines how frequently such events occur.

3.4 Analytical Constraint Screening

Before evaluating the authentication architectures against observed ADS-B traffic, an analytical screening is performed to quantify the communication requirements imposed by each signature scheme. The screening isolates the effects of signature size and authentication interval from traffic-dependent effects and establishes the minimum transport requirements that must subsequently be reproduced in the trace-driven evaluation. Computational performance is treated separately in section 4, since signing and verification delays depend on the implementation and hardware platform from which the benchmark values are obtained.

Let S_{sig} denote the signature size in bits and S_{frag} the number of bits available for authentication material within one authentication transmission. The number of transmissions required to convey one complete signature is then

$$n_{\text{frag}} = \left\lceil \frac{S_{\text{sig}}}{S_{\text{frag}}} \right\rceil. \quad (5)$$

This generalizes the theoretical lower bound introduced in Equation 1. Under the optimistic assumption that the complete 56-bit ME field of an additional Extended Squitter can be allocated to authentication material, $S_{\text{frag}} = 56$ bits. The resulting n_{frag} therefore represents a lower bound on the number of additional transmissions required per signature. A practical protocol extension would require part of this capacity for functions such as fragment identification, sequencing, message association, and replay protection, reducing S_{frag} and consequently increasing the required number of transmissions.

The communication burden of an authentication architecture depends jointly on this fragment count and the authentication interval k . Since one signature is generated for every k surveillance messages, the average number of additional authentication transmissions per surveillance message is

$$O_{\text{tx}} = \frac{n_{\text{frag}}}{k}. \quad (6)$$

For an aircraft generating surveillance messages at an average rate $r_{\text{ADS-B}}$, the corresponding authentication transmission rate is

$$r_{\text{auth}} = r_{\text{ADS-B}} \frac{n_{\text{frag}}}{k}. \quad (7)$$

Assuming an Extended Squitter airtime of t_{ES} , the additional channel occupancy introduced by the authentication traffic of one aircraft is

$$C_{\text{auth}} = r_{\text{auth}} t_{\text{ES}} = r_{\text{ADS-B}} \frac{n_{\text{frag}}}{k} t_{\text{ES}}. \quad (8)$$

These relationships expose the principal communication trade-off independently of a particular traffic trace. Signature size determines the number of transmissions required for each authentication event, while k determines how frequently those events occur. Increasing the authentication interval therefore amortizes the communication cost of a signature across a larger number of surveillance messages, but does not reduce the number of fragments required to reconstruct an individual signature. Conversely, reducing signature size directly reduces the transport requirement for every authentication event and is therefore beneficial under both immediate and detached authentication.

Fragmentation also affects the probability that an authentication event can be completed successfully. If the reception of each authentication fragment is represented by an independent probability p_{rx} , the probability of reconstructing a complete signature from n_{frag} required fragments is

$$P_{\text{complete}} = p_{\text{rx}}^{n_{\text{frag}}}, \quad (9)$$

as introduced in Equation 2. This expression is retained only as an analytical illustration of the sensitivity of fragmented authentication to message loss, and is not used as a substitute for the reception behavior represented by the trace-driven evaluation. In particular, real 1090 MHz reception is not characterized by independent and identically distributed fragment losses.

The analytical screening therefore establishes three quantities for each combination of signature scheme and authentication interval: the minimum fragment count per signature, the average number of additional authentication transmissions per surveillance message, and the corresponding authentication-transmission rate and channel occupancy for a given ADS-B message rate. These quantities provide the deterministic communication baseline against which the OpenSky-derived

traffic scenarios are evaluated in section 4. The trace-driven analysis subsequently replaces the assumed average surveillance rate with the observed temporal distribution of ADS-B messages and incorporates signature reconstruction, authentication latency, and computational signing and verification requirements.

4. Trace-Driven Experimental Methodology

4.1 OpenSky Dataset

The trace-driven evaluation is based on historical ADS-B observations obtained from the OpenSky Network, a crowdsourced air-traffic surveillance network that collects Mode S and ADS-B transmissions through a geographically distributed infrastructure of ground-based receivers [Schafer2014OpenSky, Strohmeier2014OpenSky]. Using observed traffic rather than a synthetic message-generation model preserves the temporal characteristics of operational ADS-B transmissions, including variations in per-aircraft transmission rates, aircraft density, and the simultaneous activity of multiple aircraft. The resulting traces provide the empirical traffic input over which the hypothetical authentication mechanisms are evaluated; no cryptographic authentication material is present in the original OpenSky observations.

The dataset considered in this study comprises ADS-B 1090ES messages observed within [GEOGRAPHIC REGION] during [OBSERVATION PERIOD]. Raw message-level observations are retained with their reception timestamps and ICAO 24-bit aircraft addresses, allowing transmissions to be ordered independently for each aircraft while preserving their temporal position within the overall traffic trace. Only messages attributable to ADS-B Extended Squitter transmissions are retained for the authentication experiment. Observations lacking the information required to associate a transmission with an aircraft or to reconstruct the message input used by the simulator are excluded during preprocessing. No synthetic surveillance messages are introduced into the baseline trace.

Following preprocessing, the dataset contains [N] ADS-B messages associated with [N] unique aircraft over an observation period of [DURATION]. In addition to the complete dataset, traffic intensity is characterized over time to preserve variations in the number of simultaneously active aircraft and observed message rates. These variations are particularly relevant to the evaluation because authentication traffic is superimposed on the observed surveillance transmissions: the communication cost of a given authentication configuration therefore depends not only on the size of an individual signature, but also on when and how frequently aircraft generate messages within the trace. The principal characteristics of the resulting dataset are summarized in Table 2.

Table 2. Characteristics of the OpenSky dataset used for the trace-driven evaluation.

Characteristic	Value
Observation date(s)	[TODO]
Observation period	[TODO]
Geographic region	[TODO]
Trace duration	[TODO]
ADS-B messages	[TODO]
Unique aircraft	[TODO]
Mean message rate	[TODO]
Peak active aircraft	[TODO]

The OpenSky observations are used exclusively to reproduce the timing and density of the underlying surveillance traffic. Authentication signatures, their fragmentation, additional transmissions,

cryptographic processing delays, and verification events are generated by the replay simulator described in subsection 4.2. This separation ensures that the baseline traffic remains trace-derived while the effects of the evaluated authentication configurations are introduced in a controlled and reproducible manner.

4.2 Replay Simulator 319

4.3 Computational Model 320

4.4 Experimental Variables 321

4.5 Evaluation Metrics 322

4.6 Reproducibility 323

5. Results 324

5.1 Analytical Screening Results 325

5.2 Baseline OpenSky Traffic Characteristics 326

5.3 Communication Overhead and Channel Occupancy 327

5.4 Authentication Latency 328

5.5 Authentication Reconstruction and Success 329

5.6 Verification Workload 330

5.7 Sensitivity Analysis 331

6. Discussion 332

6.1 Limits of Authentication 333

6.2 Algorithm and Architecture Trade-offs 334

6.3 Future CNS Evolution 335

6.4 Limitations 336

7. Conclusion 337

References 338

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426
427